

Wei Chen

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Education

University of Wisconsin–Madison

Aug 2024 – Dec 2025

M.S. in Electrical and Computer Engineering
Machine Learning and Signal Processing (MLSP), GPA: 3.73 / 4.00

Sichuan University

Sep 2020 – Jun 2024

B.S. in Flight Vehicle Control and Information Engineering
GPA: 3.79 / 4.00

Selected Projects

My Transformer

[GitHub](#)

- Built a from-scratch PyTorch encoder-decoder Transformer for machine translation, including multi-head attention, positional encoding, training, inference, and evaluation.
- Implemented data preparation, greedy / beam-search decoding, BLEU evaluation, checkpointing, testing, and attention visualization.

Global-Market-Newsboard

[GitHub](#)

- Developed an open-source market news dashboard focused on turning real-time information streams into a more usable interface for retail traders.
- Built a user-facing workflow spanning news aggregation, data handling, presentation logic, and practical decision-support design; deployed at globalnewsboard.cn (demo password: 123456).

Campus Lane Detection

[GitHub](#)

- Built a course project around ultra-fast deep lane detection in PyTorch using a structure-aware lane detection pipeline with training, evaluation, visualization, and speed-testing scripts.
- Adapted TuSimple / CULane-style tooling to campus driving footage and produced a 1080p / 30fps curved-lane overlay demo.

Research Manuscripts

- *Basis-Mediated Bilinear Attention: A New Method for Greatly Reducing Query–Key Pathway Parameters*. Submitted to NeurIPS 2026. Proposes BMB/BMB-UV for query-key pathway reparameterization with shared latent bases; evaluated on ViT-Base ImageNet-1K and GPT-2 C4 pretraining.
- *Revisiting Positional Priors in Vision Transformers with Semantic Gating*. Manuscript in preparation on semantic gating beyond static geometry-driven bias in ViT attention.
- *Beyond Positional Encoding: Layerwise Attention Reconstruction via Query-Key Structural Priors*. Manuscript in preparation on layerwise attention reconstruction using query-key structural priors.

Skills

- **LLM Pretraining Systems**: distributed pretraining, throughput profiling, checkpointing, mixed precision, PyTorch
- **Memory and Efficiency**: activation checkpointing, FlashAttention, KV-cache optimization, quantization, PEFT, LoRA, DoRA, QLoRA
- **Optimization and Scaling**: Adam, AdamW, Lion, Shampoo, SOAP, Muon; scaling laws, data recipes, compute-data trade-offs, model size planning
- **Post-Training and Alignment**: SFT, RLHF, preference alignment, DPO-style objectives, prompt optimization, instruction data refinement
- **Agents and Coding Tools**: ReAct, tool use, multi-agent workflows, Codex, Claude Code, Kimi, Swarm, DSPy, GEPA

Selected Coursework

- Large Language Models and Deep Learning: [Advanced Topics in Large Language Models \(ECE 901\)](#); [Introduction to Artificial Neural Networks \(CS 539\)](#)
- Machine Learning and Optimization: [Machine Learning \(ECE 760\)](#); [Matrix Methods in Machine Learning \(ECE 532\)](#); [Topics in Mathematical Data Science \(ECE 888\)](#)
- Perception and Signal Processing: [Image Processing \(ECE 533\)](#)